Campus City Logistics Optimization

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1 Introduction

Efficient logistics planning is essential for minimizing operational costs while ensuring timely delivery of essential resources. Large campuses require a reliable distribution system for delivering supplies to facilities such as hospitals, dormitories, libraries, and academic buildings.

This project focuses on designing an optimized logistics distribution network that minimizes annual logistics cost while satisfying operational constraints.

2 Problem Statement

Campus City Logistics currently operates with an inefficient supply distribution system. The objective of this project is to determine the optimal warehouse locations and distribution plan that minimizes total annual cost while ensuring that all facilities receive their required supply.

The optimization model must satisfy the following constraints:

- Each facility must receive its required demand.
- Warehouse capacity limits must not be exceeded.
- Exactly two warehouses must be selected for redundancy.
- The total annual cost must remain within the allocated budget.

3 Dataset Description

The project uses the following datasets:

- Facilities dataset containing facility locations and types
- Demand dataset containing daily supply requirements
- Warehouse dataset containing capacity and cost information
- Transportation cost matrix between warehouses and facilities
- Geographic boundary data defining the campus region

4 Mathematical Model

The problem is formulated as a Mixed Integer Linear Programming (MILP) model.

4.1 Decision Variables

- y_w : Binary variable indicating whether warehouse w is selected.
- x_{wf} : Number of units shipped from warehouse w to facility f.

4.2 Objective Function

The objective is to minimize total annual cost:

$$\text{Minimize } \sum_{w,f} C_{wf} x_{wf} + \sum_{w} (C_w^c + C_w^o) y_w$$

Where:

- C_{wf} is transportation cost per unit
- C_w^c is annualized construction cost
- C_w^o is annual operational cost

4.3 Constraints

Warehouse Selection

$$\sum_{w} y_w = 2$$

Demand Satisfaction

$$\sum_{w} x_{wf} = D_f$$

Warehouse Capacity

$$\sum_{f} x_{wf} \leq Cap_w$$

Budget Constraint

$$Total\ Cost \leq 1{,}500{,}000$$

5 Implementation

The optimization model was implemented in Python using the following libraries:

- PuLP for solving the MILP optimization problem
- Pandas for dataset processing
- Folium for geographic visualization
- Plotly for data visualization

The system reads the datasets, constructs the optimization model, solves the problem using a linear programming solver, and generates visualizations for analysis.

6 Results and Visualization

6.1 Cost Breakdown

The following figure illustrates the distribution of annual logistics costs among transportation, construction, and operational costs.

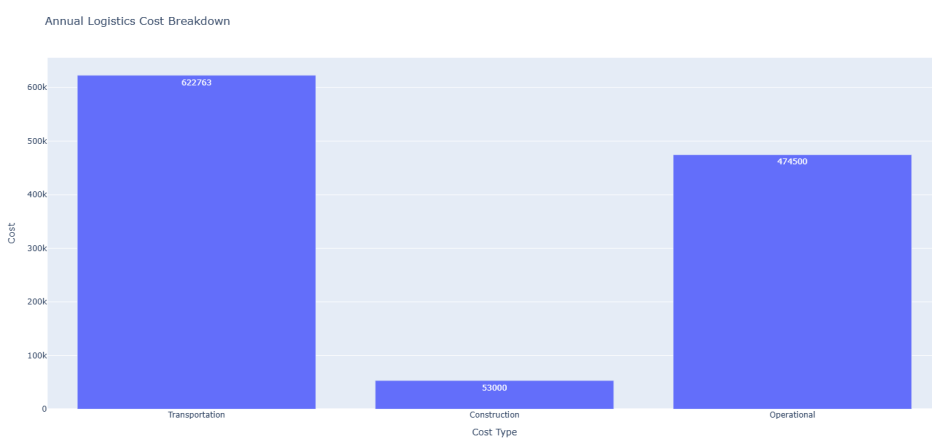

Figure 1: Annual Logistics Cost Breakdown

6.2 Shipment Distribution

The heatmap below shows the shipment distribution from warehouses to facilities.

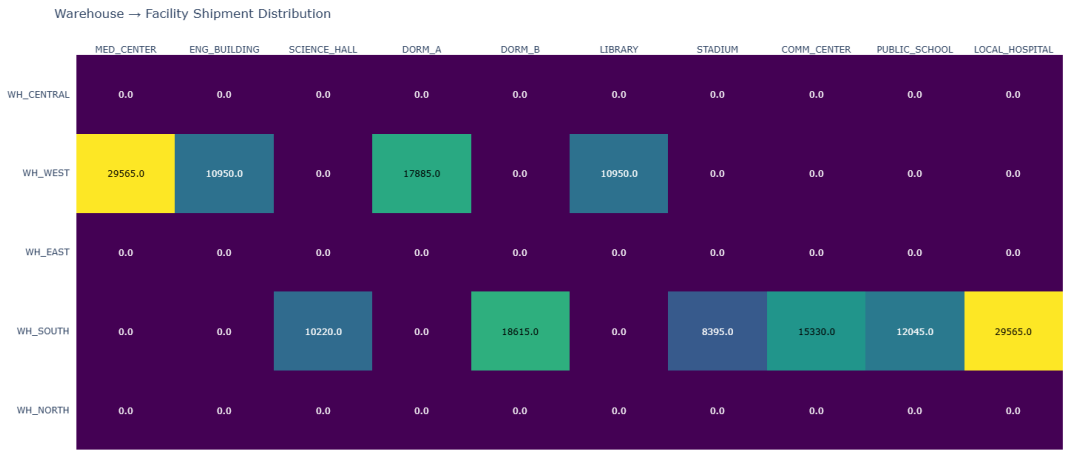

Figure 2: Warehouse to Facility Shipment Distribution

6.3 Optimized Logistics Network Map

The logistics network map visualizes facility locations, warehouse locations, and shipment routes generated by the optimization model.

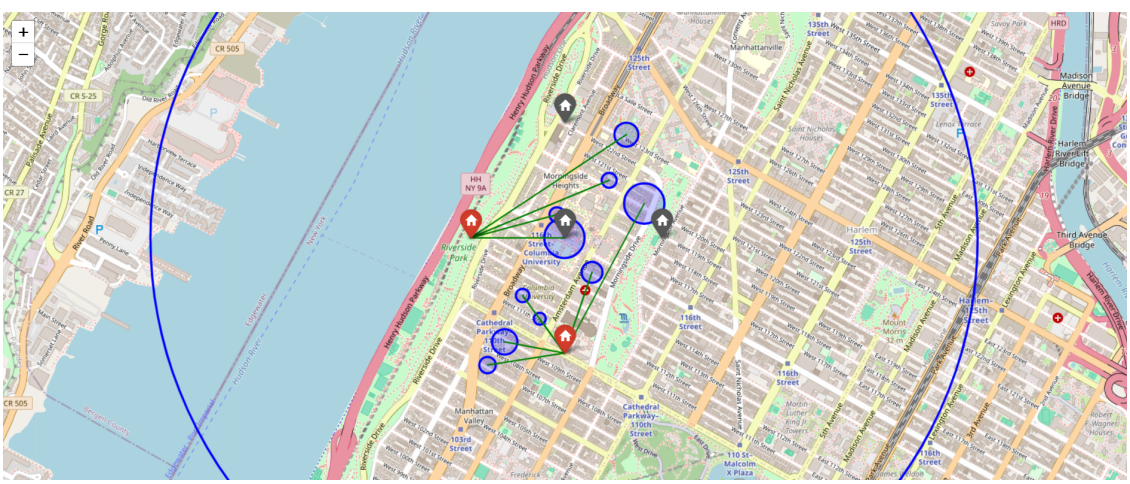

Figure 3: Optimized Campus Logistics Network

7 Discussion

The results show that selecting strategically located warehouses significantly reduces transportation costs while maintaining operational efficiency. Transportation cost represents the largest portion of the logistics expenditure, indicating the importance of efficient facility-to-warehouse allocation.

The shipment heatmap highlights how facilities are assigned to warehouses based on cost efficiency.

The geographic logistics network map provides a visual understanding of shipment routes and facility coverage.

8 Conclusion

This project demonstrates how Mixed Integer Linear Programming can be applied to optimize logistics networks. By combining optimization algorithms with geographic visualization tools, the system provides an effective solution for planning supply distribution across campus facilities.

The approach reduces operational costs and improves resource allocation efficiency.